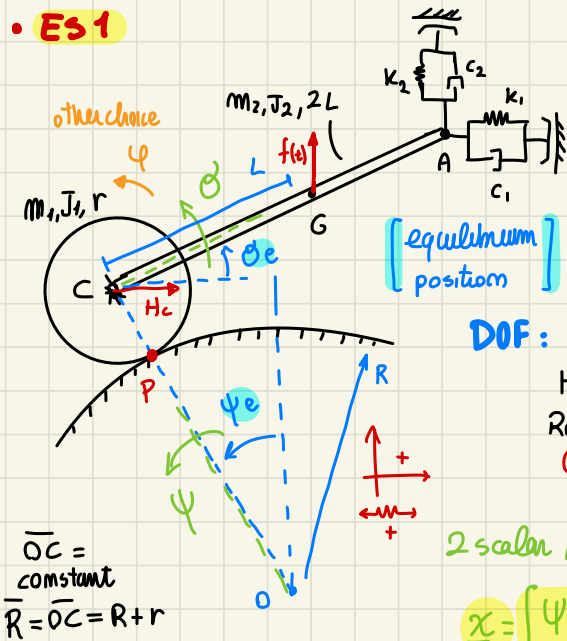




ESE 9 CONSTRAINT FORCES, SPRING PRE-LOAD

• ES1



(PROBLEM from a written exam)

2 RIGID BODIES { Disk hinged on C
 Rigid BAR

disk roll w/o sliding over a circular shape guide of radius R

{ K_1 : HORIZ. of A
 K_2 : VERTIC. of A }

DOF: $2 \times 3 = 6$ DOF

HINGE(C) -2 DOF

Roll. without slid. -2 DOF

(Pathway $\psi = \theta$
 and disk on contact)

Independently on the kind of guide

2 DOF

2 scalar indep coordinates?

$\underline{x} = \begin{bmatrix} \psi \\ \theta \end{bmatrix}$ ← Rotation from θ_e

We need coord
 to identify the

disk on the guide $\Rightarrow$ we have 2 choices!

$\psi = \theta \rightarrow$ when $\psi_e, \theta \rightarrow 0$

OC rotate! there are PRO/CONS

on this choice! $\rightarrow$ NOTICE we are

taking a coord. NOT strictly part of the
 system body rotation... IT describe system motion

KINETIC EN.

$$T = \frac{1}{2} \dot{\underline{y}}_m^T [m_{ph}] \dot{\underline{y}}_m \quad \underline{\dot{y}}_m = \begin{bmatrix} v_{Cx} \\ v_{Cy} \\ \omega_1 \\ v_{Gx} \\ v_{Gy} \\ \omega_2 \end{bmatrix}$$

$$[m_{ph}] = \text{diag}\{m_1, m_1, J_1, m_2, m_2, J_2\}$$

We can write
 JACOBIANS from a

NON linear expression of G, C

and get all the terms of $[\lambda_m]$

$\hookrightarrow$

$$\begin{cases}
 x_c = -\bar{R} \sin(\psi + \psi_e) \approx -\bar{R} \cos(\psi_e) \psi - \bar{R} \sin(\psi_e) \left(\cancel{1 - \psi^2/2} \right) \\
 y_c = +\bar{R} \cos(\psi + \psi_e) \approx \bar{R} \cos(\psi_e) \left(\cancel{1 - \frac{\psi^2}{2}} \right) - \bar{R} \sin(\psi_e) \psi
 \end{cases}$$

CONST.

$$\begin{cases}
 x_G = x_c + L \cos(\theta_e + \theta) = -\bar{R} \cos(\psi_e) \psi + \bar{R} \sin(\psi_e) \psi^2/2 + L \cos(\theta_e) \left(\cancel{1 - \theta^2/2} \right) \\
 \quad \quad \quad - L \sin(\theta_e) \theta \\
 y_G = y_c + L \sin(\theta_e + \theta) = -\bar{R} \cos(\psi_e) \frac{\psi^2}{2} - \bar{R} \sin(\psi_e) \psi + L \cos(\theta_e) \theta \\
 \quad \quad \quad + L \sin(\theta_e) \left(\cancel{1 - \theta^2/2} \right)
 \end{cases}$$

When neglecting carefully of product (only constants!)

from this relationship, easily

$$[\lambda_m] = \begin{bmatrix} \dot{\psi} & \dot{\theta} \\ -\bar{R} \cos(\psi_e) & 0 \\ -\bar{R} \sin(\psi_e) & 0 \\ \bar{R}/r & 0 \\ -\bar{R} \cos(\psi_e) & -L \sin(\theta_e) \\ -\bar{R} \sin(\psi_e) & L \cos(\theta_e) \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_{cx} \\ V_{cy} \\ \omega_1 \\ V_{Gx} \\ V_{Gy} \\ \omega_2 \end{bmatrix}$$

from time derivative of

$$\dot{x}_c, \dot{y}_c \rightarrow V_{cx}, V_{cy}$$

look @ I ORD terms

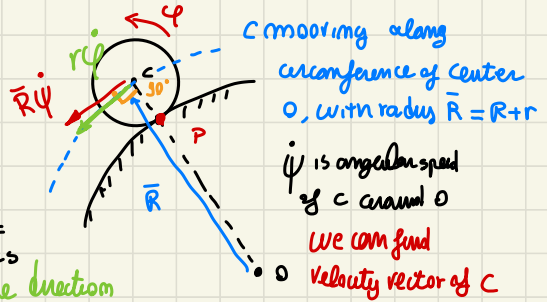
NOT consider the II ORD terms on the Jacobian

$\omega_1 \leftarrow$ carefully here $\dot{\psi} \neq \omega_1$

$\omega_1 = \dot{\psi}$ ROTATION

we should connect $\dot{\psi}, \dot{\theta}$!
RELATIONSHIP

where $\dot{\theta}$ does NOT influence π



$$r \dot{\psi} = \bar{R} \dot{\psi}$$

$$\text{so } \dot{\psi} = \frac{\bar{R}}{r} \dot{\psi}$$

BIG ratio depending on the problem geometry

canceling P the C.I.R of the disk

We can write ω_c respect its C.I.R as

$\dot{\psi} \cdot r$ same direction with magnitude $\dot{\psi} r$

same reasoning to find V_{Gx}, V_{Gy} highlighting I ORD TERMS

POTENTIAL $\Delta \ell_1 = -X_A$ for usual sign convention $\xleftarrow{(+)} \text{---} \xrightarrow{(+)}$ if elongate
 ($X_A > 0$ means a compression)

it is easy to find X_A from X_G JUST using $2L$ instead of L

$$\begin{cases} \Delta \ell_1 = + \bar{R} \cos(\psi_e) \psi - \bar{R} \sin(\psi_e) \frac{\psi^2}{2} + 2L \cos(\theta_e) \frac{\theta^2}{2} + 2L \sin(\theta_e) \theta \\ \Delta \ell_2 = -y_A = \bar{R} \cos(\psi_e) \frac{\psi^2}{2} + \bar{R} \sin(\psi_e) \psi - 2L \cos(\theta_e) \theta + 2L \sin(\theta_e) \frac{\theta^2}{2} \end{cases}$$

↑
from y_G but with $2L$
 ↓ we can form

II ORD

$$[\lambda_K] = \begin{bmatrix} +\bar{R} \cos(\psi_e) & 2L \sin(\theta_e) \\ -\bar{R} \sin(\psi_e) & -2L \cos(\theta_e) \end{bmatrix} \begin{matrix} \Delta \ell_1 \\ \Delta \ell_2 \end{matrix}$$

JUST using
 linear relation (I ORD)

- FRF
 - Re compute static eq. pos if change data
- addit. question later

$$[\lambda_F] = \begin{bmatrix} -\bar{R} \sin(\psi_e) & L \cos(\theta_e) \end{bmatrix} \delta y_c$$

same relationship
 as V_{cy} row om $[\lambda_m]$

at end to find also II ORD $[K_y]$, $[K_{el II}]$

$$[K_{el II,1}] = K_1 \Delta \ell_{o1} \begin{bmatrix} -\bar{R} \sin(\psi_e) & 0 \\ 0 & 2L \cos(\theta_e) \end{bmatrix}$$

$\frac{\partial^2 \Delta \ell_1}{\partial \theta^2}$ and so on respect ψ, θ
 Looking II ORD terms on $\Delta \ell_1, \Delta \ell_2$

$$[K_{el II,2}] = K_2 \Delta \ell_{o2} \begin{bmatrix} \bar{R} \cos(\psi_e) & 0 \\ 0 & 2L \sin(\theta_e) \end{bmatrix}$$

No mix terms as ψ, θ
 effect on $\Delta \ell$!

it remains to find $[K_{g1}]$, $[K_{g2}]$

from y_c expression

from y_c expression

looking II ORD parts

$$[K_{g1}] = m_1 g \begin{bmatrix} -\bar{R} \cos(\psi_e) & 0 \\ 0 & 0 \end{bmatrix}$$

$$[K_{g2}] = m_2 g \begin{bmatrix} -\bar{R} \cos(\psi_e) & 0 \\ 0 & -L \sin(\theta_e) \end{bmatrix}$$

so $[K] = [\lambda_k]^T [K_{ph}] [\lambda_k] + [K_{g1}] + [K_{g2}] + [K_{elI,1}] + [K_{elI,2}]$
(do with MATLAB)

easy way same to write $[C] = [\lambda_c]^T [C_{ph}] [\lambda_c]$

same as $[\lambda_k]$ but with C_1, C_2

so easily you

find all MATRICES $[M], [C], [K] \rightarrow$ from here take **MATLAB**

$\Leftarrow \text{eig}([M]^{-1}[K])$ obtain $\omega^2, \underline{x}$

NAT FREQ. and MODAL shape vector

deriving some **FRF**
important of the problem

you can easily get FRF respect ψ, θ when $f(t)=1$ acts harmonic with Ω changes

build FRF for some relevant point of displacement y_c ? $\Rightarrow$ FRF of y_c

aiming to find LINEARIZED I/O relationship

once found FRF for $\underline{x}$, so ψ, θ will

connect linearly ψ, θ to y_c

easy to take from $[\lambda_m]$ row of $[V_{ay}]$

from FRF of $\underline{x} \Rightarrow y_c = -\bar{R} \sin(\psi_e) \underline{\chi(1)} + L \cos(\theta_e) \underline{\chi(2)}$

complex value,
we will extract

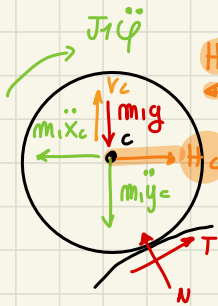
$\begin{cases} \text{amplitude } (y_c) \\ \text{phase } (y_c) \end{cases}$ as VECTOR of 1-dimensional

Complex value, with harmonic input evaluated for Ω values $\in [\text{range}]$ iterating once for cycle

• QUESTION: for some $f_c(z)$ compute H_c
(horizontal constrain on c)

looking the HINGE (c) CONSTRAINT FORCE component easier to compute by D'Alemb.
"opening" the hinge

$m_1 g$ NOT RELEVANT on FRF(*)



Reaction forces exchanged with the bar
reaction forces of the GUIDE (NORMAL + TANGENTIAL)

JUST LOOKING DISK
IS NOT ENOUGH...

(4 equations on same body are totally un-useful! JUST 3 is enough!)
4TH is linear combination of the 3 important

you have to look also on the BAR!

H_c, V_c as on DISK but reverse direction (III NEWTON)

carefully on BODE PLOT sign convention respect the choice done...

$f_0=1$ is computing FRF

from [14]

$$F_{el2dyn} = (K_1 + i\Omega C_1)\Delta e_1 \rightarrow \Delta e_1 = \bar{R} \cos(\psi_e) X(1) + 2L \sin(\psi_e) X(2)$$

We can compute H_c JUST from 1 equation:

$$\sum_{i \in \text{BAR}} F_{x_i} = 0 \quad \text{forces on the BAR}$$

$$\sum_{\text{BAR}} F_x = 0 :$$

$$H_c = -m_2 \ddot{x}_G + F_{el1dyn}$$

We know $\ddot{x}_G = -\Omega^2 x_G =$ harmonic solution

$$= -\Omega^2 (-\bar{R} \cos(\psi_e) X(1) - L \sin(\psi_e) X(2))$$

easy to compute
on the MATLAB
script!

from [14] row of V_{x_0}

(*) FRF is related to DYNAMIC components! I/O relationship on the dynamic I and O, JUST fluctuating component of forces!

FRF of This?

here we have

4 unknown FORCE

T, N, H_c, V_c ... NOT easy to extract H_c ...

You can write 3 imp eq. for D'Alemb.
NOT enough having 4 unknn.

Kinematical relationships are easy to take from [1] Jacobians

knowing, of harmonic functions of time $\dot{x} = i\Omega x$

all to find H_2 FRF

$$\ddot{x} = -\Omega^2 x$$

x complex value!

• QUESTION? compute STATIC preload of k_1, k_2 SPRINGS when $(\Delta l_{10}, \Delta l_{20})$

how change the pre-load of the system

$\downarrow$ if $m_1 = 10 \text{ kg}$ instead of $m_1 = 5 \text{ kg}$ previously (we had $\Delta l_{10}, \Delta l_{20}$ consistent with initial data)

changes... Assuming same ψ_e, δ_e equilibrium position NOT changing but different m_1 ... How does $\Delta l_{10}, \Delta l_{20}$ changes? (conserving same equilibrium) on New Syst.

STATIC EQUILIBRIUM PROBLEM!

(NOTICE)

be consistent when drawing elastic forces for Δl_1 the corresponding F_{el1}

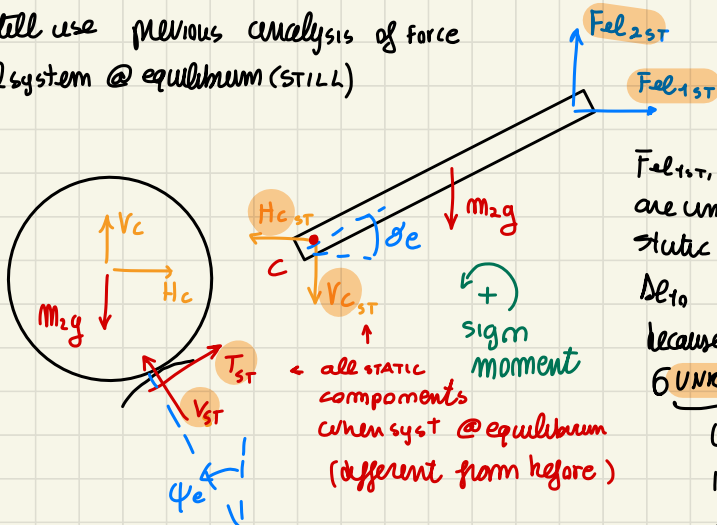
reason considering tension spring and from here draw F_{el} when spring in tension effect

we are talking **STATIC CONDITION**

any force depending on vel, acc goes to 0 $\dot{x}=0$
 $\ddot{x}=0$

so the damper have NO effect, also INERTIA... and neglect also $f(t)$ JUST weight + STATIC forces! (the one neglected on FRF solving)

We still use previous analysis of force all system @ equilibrium (STILL)



F_{el1st}, F_{el2st} are unknown because static condition $\Delta l_{10}, \Delta l_{20}$ has changed! because m_1 changes
6 UNKNOWN

We just are interested on $F_{el1st} \rightarrow \Delta l_{10} = \frac{F_{el1st}}{k_1}$
 $F_{el2st} \rightarrow \Delta l_{20} = \frac{F_{el2st}}{k_2}$

taking a general "approach" we write uncomfortable 6 eq in 6 unk
 you can solve it in an easier way...
 (different approach)

if we can't find JUST 2 eq. $\begin{cases} F_{el1} \\ F_{el2} \end{cases}$

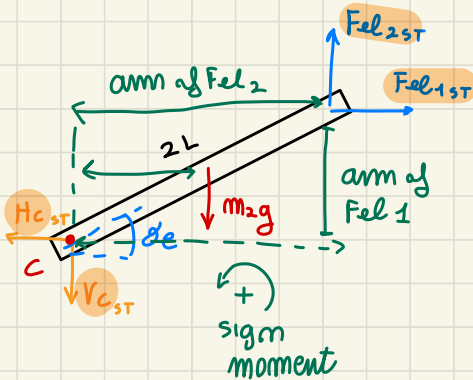
we want 2 eq respect only (F_{el1}, F_{el2}) for each

to eliminate unknowns on the point C of BAR:

moment of all forces acty on BAR centered on C

(easy way to neglect all forces acty on a point, having arm 0)

$$\sum_{\text{BAR}} \vec{M}_C = 0 \quad (\text{only function of } F_{el1}, F_{el2} \text{ unk.}) - F_{el1st} \cdot 2L \sin(\vartheta_e) + F_{el2st} \cdot 2L \cos(\vartheta_e) = m_2 g L \cos(\vartheta_e)$$



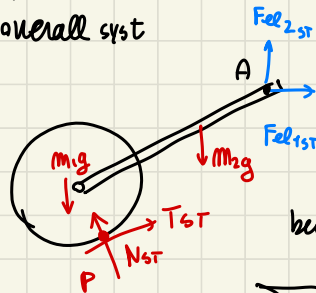
involve only unknowns wanted

If eq is harder to find...

consider the ENTIRE system
 of the two bodies, the internal
 constraint becomes internal
 forces to neglect (exchange)

we consider the overall syst
 H_{cst} cancel between DISK and BAR, same for V_{cst}
 So H_c, V_c does NOT appear

$\curvearrowright +$



still $N_{ST}, T_{ST} \neq 0$ because
 exchanged with the guide of disk
 but easily writing

hard ARM
 calculation...

$F_{el1} \curvearrowright (-)$ effect

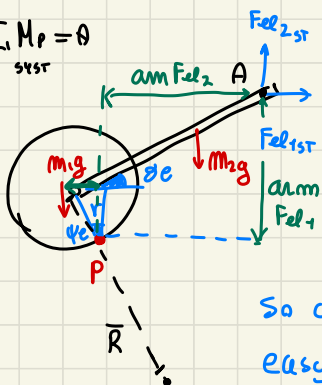
$F_{el2} \curvearrowright (+)$

$m_1 g \curvearrowright (+)$ $m_2 g \curvearrowright (-)$

$$\sum_{\text{SYST}} \vec{M}_P = 0$$

cancel each
 unknowns!
 JUST equation
 of F_{el1}, F_{el2}

$$\sum M_P = 0$$



$$\begin{aligned}
 & -F_{el1st} (r \cos(\psi_e) + 2L \sin(\theta_e)) + \\
 & + F_{el2st} (-\bar{r} \sin(\psi_e) + 2L \cos(\theta_e)) \\
 & = -m_1 g r \sin(\psi_e) + m_2 g (-\bar{r} \sin(\psi_e) + L \cos(\theta_e))
 \end{aligned}$$

2nd eq.

So we have 2 eq in 2 unk F_{el1st} , F_{el2st}
 easy to solve as MATRIX form square coefficients
 form of writing it

$$[A] \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b \\ c \end{pmatrix}$$

coeff of F_{el1}, F_{el2} " expressions on right

$\begin{pmatrix} F_{el1st} \\ F_{el2st} \end{pmatrix} \rightarrow$ and from here Δb_{10}
 Δb_{20} directly found!

- Some problem can be solved Imposed **STATIONARITY** of system
 Writing V overall potential energy, considering that IORD will
 cancel with each other and find pre-load from here!